

Smartphone-Based 3D Visualization for Grape Berry Thinning Assistance

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Abstract. To help novices learn how to thin grape berries, we developed an application that automates the entire workflow, from video capture to the presentation of thinning targets. The 3D position of each berry is estimated through point-cloud reconstruction followed by sphere detection. The selection of berries to be thinned is then formulated as a combinatorial optimization problem and solved using a genetic algorithm. The resulting thinning targets are visualized on an interactive 3D model, allowing users to intuitively check which berries to remove. The entire user-facing interface runs on a commodity smartphone in the field. In a pilot evaluation using real grape bunches, thinning guided by the application achieved inter-berry spacing comparable to that of an expert.

Keywords: Grape berry thinning · 3D reconstruction · Smartphone application · 3D visualization · Precision agriculture

1 Introduction

High-quality grape production requires an attractive bunch shape, which depends on *berry thinning* – removing excess berries to enlarge and sweeten the rest. Growers must quickly choose which berries to remove, balancing berry count, spacing, and growth space. Moreover, this work must be completed for all bunches within about ten days. This makes the skill hard to learn. In addition, as growers age and decline in number, transferring the skill is becoming urgent. Our goal is therefore to develop an IT-based system that supports grape berry thinning and helps novices learn it quickly.

Buayai *et al.* [1] detect berries to remove from 2D images using a deep neural network and present them via AR smart glasses. However, their method ignores the berries’ 3D arrangement and requires costly dedicated hardware.

We instead select targets from the berries’ 3D arrangement: from a video we estimate each berry’s 3D position, formulate selection as a combinatorial optimization problem, and solve it with a genetic algorithm for even distribution. Targets are shown on a 3D model on a commodity smartphone for intuitive confirmation without dedicated hardware.

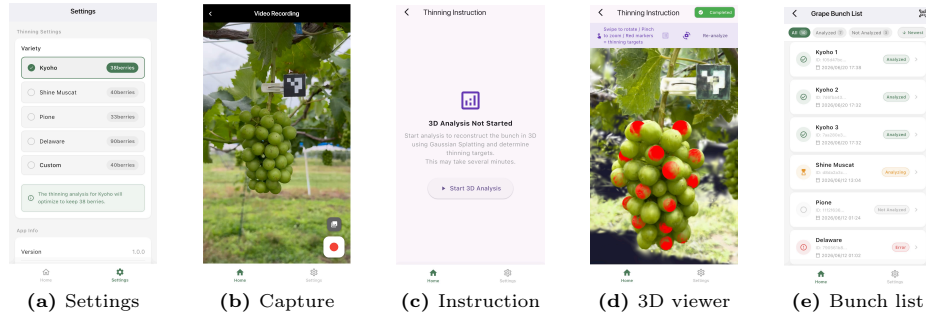


Fig. 1: Screens of the developed application

2 Proposed System

The user records one video sweep around the bunch with an ArUco marker [3] in view (Sec. 3). The server then reconstructs the bunch, detects the berries, and computes the targets, which are displayed on an interactive 3D model (Fig. 1d).

The system comprises a smartphone, a cloud-based data-management platform, and a GPU server. Because 3D reconstruction and optimization are too heavy for a phone, the smartphone handles only video capture and result display, exchanging data asynchronously with the server through the cloud platform.

The server runs four stages, following the reconstruction and selection algorithm of a related, concurrently submitted work [7]; this paper instead contributes the smartphone application, its interactive visualization, and a user study of its effect on real thinning. First, a dense 3D Gaussian point cloud is reconstructed from the multi-view video frames using Structure from Motion (SfM) [5] and Eliminating Densification for Efficient Convergence of 3DGS (EDGS) [4]. DBSCAN [2] then extracts the densest point cluster, which is treated as the grape bunch.

Second, the arbitrary reconstruction scale is converted to real-world units using an ArUco marker of known size, taking the median ratio of real to reconstructed inter-camera distances over all frame pairs.

Third, spheres are detected using Random Sample Consensus (RANSAC) [6] to estimate each berry’s 3D position and radius. Only spheres within the expected radius range and below a fitting-error threshold are kept as valid berries.

Finally, target selection is formulated as a combinatorial optimization problem and solved using a genetic algorithm. The algorithm maximizes an objective function that combines the sum of inter-berry distances and the deviation from the target berry count.

3 Application Implementation

The application has five screens, as shown in Fig. 1. On the settings screen (Fig. 1a), the user selects the cultivar and the target berry count; presets are

available for common cultivars, and other cultivars can be added as custom entries. On the capture screen (Fig. 1b), the user records or selects a video of the bunch with an ArUco marker in view; the video is then uploaded automatically. On the instruction screen (Fig. 1c), tapping “Start 3D Analysis” runs the pipeline with real-time progress. The 3D viewer (Fig. 1d) then shows the bunch with the target berries marked in red. The user can rotate and zoom the model by touch, and remove a marker with a confirming two-step tap to prevent mistakes. The bunch list screen (Fig. 1e) manages all bunches and supports filtering, sorting, and per-bunch notes.

4 Experiments

We evaluated the accuracy of the reconstruction and detection, then the application’s effectiveness in real thinning, using a Flutter-based app (iPhone 15 Pro), a Firebase backend, and an NVIDIA H100 PCIe GPU server.

4.1 Accuracy of Reconstruction and Berry Detection

We evaluated berry-count accuracy on 10 Shine Muscat bunches with berries approximately 10 mm in diameter, captured in a field. Of these, eight were reconstructed successfully; the other two failed due to motion blur and occlusion. Ground truth counts were obtained manually by one author. Detected counts correlated strongly with the ground truth (Pearson $r = .985$, $p < .001$), with no significant bias (t-test/Wilcoxon, $p = .41/.52$), and the Bland–Altman 95% limits of agreement were about ± 4 berries. Detailed data are provided in the supplementary material.

4.2 Effectiveness of the Berry-Thinning Support Application

We conducted the experiment on June 11, 2026, at the Osaka Metropolitan University research field, using Kyoho bunches with a target of 38 berries. Three participants each thinned three different bunches (nine in total): one novice using the application (Application), one novice with no experience (Novice), and one expert (Expert). The RANSAC radius search range was 4–10 mm. The Application participant followed the 3D viewer’s targets, while the Novice and Expert were simply told to leave 38 berries. We measured the working time, then recorded and processed each bunch with the pipeline to obtain the remaining berries’ 3D coordinates.

We use three metrics: accuracy (error from the target count of 38), efficiency (average time per berry), and spatial uniformity (mean nearest-neighbor (NN) distance and its standard deviation, with bunch length normalized to the nine-bunch mean). A larger NN distance means a more even, less dense arrangement, and a smaller standard deviation means more uniform spacing.

Table 1 shows the per-condition averages. Application matched Expert in mean NN distance (Tab. 1) because its targets maximize the sum of inter-berry

Table 1: Experimental results (per-condition averages). “Before”/“After”: berry count before/after thinning; “|Err.|”: absolute error from the target count (38); “s/berry”: average time to thin one berry; “NN”: NN distance between berries.

Participant	Berry count			Efficiency		Spatial distribution (mm)		
	Before	After	Err.	Time (m:s)	s/berry	Length NN	mean	NN std
Application	67.3	38.0	2.0	11:33	23.6	142.0	20.8	2.8
Novice	61.0	37.0	2.3	9:02	22.6	124.6	17.6	2.0
Expert	54.3	38.0	1.3	6:57	25.5	139.8	20.7	2.1

distances, producing an even arrangement. The absolute error from the target of 38 berries was 1.3 for Expert, 2.0 for Application, and 2.3 for Novice. Application outperformed Novice, though its remaining error stems from analysis errors and worker mistakes. The per-berry time was shortest for Novice, then Application, then Expert. For Application, working time covered only the thinning step after targets appeared on screen; capture and server-side computation (395.2 ± 58.8 s per bunch) were excluded, as they can run in the background while the user handles other bunches. Even so, the application gave no time reduction, because the cognitive load of spatial alignment between the 3D targets and the real bunch offsets the time saved by reducing decision hesitation.

Because each condition had only one participant ($n = 3$ bunches), we used nonparametric tests. Exact-permutation Kruskal–Wallis tests found no significant difference for any metric, with only a weak trend for the mean NN distance ($p = .07$). A TOST equivalence test on that metric (± 2.0 mm) gave $p = .099$. Effect sizes showed that the Application–Expert pair almost fully overlapped (Cliff’s $\delta = -0.11$), whereas Application–Novice and Novice–Expert separated completely ($\delta = \pm 1.00$). These results should therefore be interpreted as preliminary trends from a pilot study; moreover, the differences in absolute count error relative to Expert (0.7–1.0 berries) are smaller than the roughly ± 4 -berry detection uncertainty reported in Section 4.1, so detection noise may also contribute to these results.

5 Conclusion

We developed a smartphone application that presents 3D-based grape berry thinning targets from video. In a pilot evaluation, Application was comparable to Expert in spatial uniformity and outperformed Novice in target-count error. Future work includes handling motion blur and occlusion, automatic 3D-to-bunch registration, a larger user study across experience levels, and a comparison with alternative guidance such as AR overlays.

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